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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S7 (R, S) / S7 (PT) (R) Examination December 2023 (2019 Scheme)

Course Code: EET401**Course Name: ADVANCED CONTROL SYSTEMS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

- 1 Define phase variables? List the advantages of choosing phase variables for state space modelling. (3)
- 2 Construct the state model for a system characterized by the differential equation (3)

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 6y + u = 0.$$
- 3 Determine the transfer function of the system with state model (3)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u ; \quad y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$
- 4 A linear time-invariant system is characterized by homogeneous state equation (3)

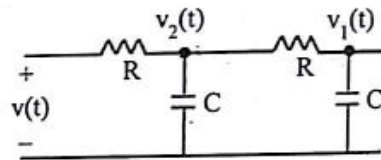
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$
 Compute the solution of the homogeneous equation, assuming the initial state vector, $X_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.
- 5 Define controllability and explain Kalman's method for determining controllability. (3)
- 6 Explain the significance of PBH test for observability. (3)
- 7 Explain the different non linearities with diagram. (3)
- 8 What is limit cycle? How will you determine stable and unstable limit cycle using phase portrait? (3)
- 9 Define singular points? Explain the types of singular points. (3)
 State the condition/s for a scalar function $V(x)$ to be (i) positive definite (ii) negative definite and (iii) indefinite. Give one example for each. (3)
- 10

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

- 11 a) From the basics, derive the state model of a Field controlled dc servo motor and draw the block diagram. (7)
- b) Develop the state model of the electrical network shown in figure by choosing $V_1(t)$ and $V_2(t)$ as state variables. (7)



OR

- 12 a) A linear time invariant system is described by the following state model

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} [u] \text{ and } y = [1 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$
 Transform this state model into a canonical state model. (10)
- b) Summarize the advantages of state space method over transfer function approach. (4)

Module II

- 13 a) A discrete time system has the transfer function

$$\frac{Y(Z)}{U(Z)} = \frac{4Z^3 - 12Z^2 + 13Z - 7}{(Z - 1)^2(Z - 2)}$$
 (7)
- Determine the state model of the system in Jordan Canonical form.
- b) The system matrix A of a discrete system is given by $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$. Compute the state transition matrix A^k using Cayley-Hamilton theorem. (7)

OR

- 14 a) For a system represented by state equation
 $\dot{X}(t) = AX(t)$, The response is $X(t) = \begin{bmatrix} e^{-2t} \\ -2e^{-2t} \end{bmatrix}$ when $X(0) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $X(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$ when $X(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Determine the system matrix A and the state transition matrix. (10)
- b) Show that eigen values of state models are unique. (4)

Module III

- 15 a) Describe about state observers. (4)
 b) Consider a linear system described by the transfer function (10)

$$\frac{Y(s)}{U(s)} = \frac{10}{s(s+1)(s+2)}$$

Design a feedback controller with a state feedback so that the closed loop poles are placed at -2, (-1+j1) and (-1-j1).

OR

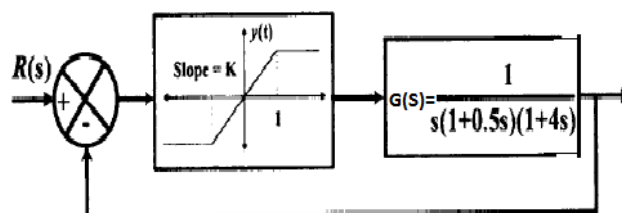
- 16 a) Consider the system described by the state model $\dot{X} = AX$, $y = CX$ where $A = \begin{bmatrix} -1 & 1 \\ 1 & -2 \end{bmatrix}$; $C = [1 \quad 0]$. Design a full-order state observer. The desired eigen values for the observer matrix are $\mu_1 = -5$; $\mu_2 = -5$; (10)
 b) Explain the concept of duality referred to controllability. (4)

Module IV

- 17 a) Develop the describing function of relay nonlinearity. (6)
 b) Define Describing function. Explain how describing function can be used for stability analysis of nonlinear systems. (8)

OR

- 18 a) Determine the value of K for an occurrence of limit cycle. Also determine the frequency, amplitude and stability of limit cycle. (10)



- b) List any four properties of non linear systems. (4)

Module V

- 19 a) A linear second order system is described by the equation (10)
 $\ddot{e} + 2\delta\omega_n\dot{e} + \omega_n^2 e = 0$, Where $\delta = 0.15$, $\omega_n = 1\text{rad/sec}$, $e(0)=1.5$, and $\dot{e}(0) = 0$.
 Determine the singular point and state the stability by constructing the phase trajectory using the method of isoclines.

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- b) Differentiate between stable and unstable limit cycles. (4)

OR

- 20 a) Define the terms (i) stability (ii) asymptotic stability (iii) asymptotically stable in the large (iv) instability (4)

- b) A second order system is represented by $\dot{X} = AX$ where $A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$. (10)

Assuming matrix Q to be identity matrix, solve for matrix P in the equation $A^T P + PA = -Q$. Use Lyapunov theorem and determine the stability of the system. Write the Lyapunov function $V(x)$
